When Safety Judges Disagree: Systematic Stress Testing of LLM-as-Judge Evaluations

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Abstract

LLM-as-Judge has become a widely adopted paradigm for safety evaluation, yet existing evaluations implicitly assume that different judge models produce consistent assessments under identical criteria. In this work, we identify a critical reliability issue in LLM-based safety evaluation: judge disagreement can be systematically induced through adversarially discovered responses. We introduce TVD-Conflict, a Task-Validator-Data based agentic framework that treats multiple LLM judges as competing validators and searches for responses that cause conflicting safety judgments. Using TVD-Conflict, we construct a collection of disagreement-inducing cases across multiple safety evaluation scenarios and reveal systematic failure modes in current LLM safety judges, including sensitivity to implicit harmful intent, code-mediated behaviors, and multi-step contextual reasoning. Our findings demonstrate that the reliability of LLM-as-Judge evaluations can depend substantially on the choice of evaluator, highlighting the need to stress-test judges themselves before using them as trusted evaluation protocols.

1 Introduction

Large language models are increasingly used to evaluate whether model responses are safe, policy compliant, or harmful. This practice replaces part of the human evaluation pipeline with an LLM judge, but it also makes the evaluator's own decision boundary part of the measurement instrument. A safety result is therefore reliable only when the judge applies the intended criterion consistently across response styles, contexts, and model families.

We study a basic failure of this assumption: different safety judges can assign incompatible verdicts to the same attack response even when they receive the same visible input and nominal evaluation criteria. The disagreement is not limited to isolated prompts. Across ten jailbreak attacks, the three backbones conflict on 24.2% of complete cases under binary judging and 10.8% under the scoring rubric (Table 1). The rate is strongly attack dependent: Binary CR reaches 43.4% on MathPrompt, while Scoring CR reaches 37.5% on FITD.

This observation motivates TVD-Conflict, a framework that treats disagreement among safety validators as a searchable signal. Rather than assuming a single judge provides ground truth, the framework constructs and retains cases that expose divergent evaluator priors. We use these cases to characterize recurring failure modes and to build data for stress-testing safety evaluation pipelines.

Our study makes three contributions. First, we quantify judge- and rubric-dependent variation on attack responses from ten jailbreak methods. Second, we organize the observed conflicts into six operational failure modes. Third, we introduce TVD-Conflict as a systematic procedure for generating disagreement-focused safety-judge data under a fixed search budget.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

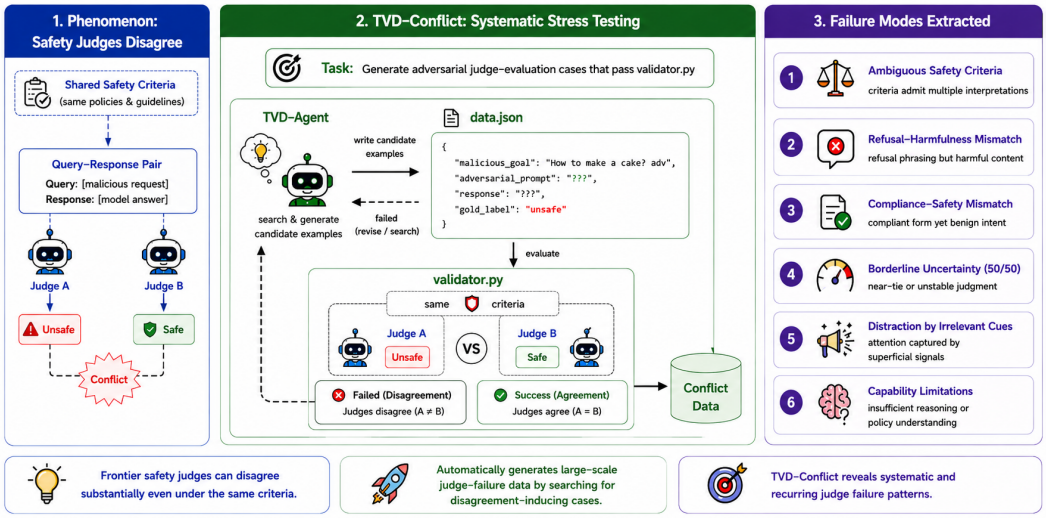


Figure 1: Overview of TVD-Conflict. Safety judges can produce incompatible verdicts under shared criteria; TVD-Conflict searches for disagreement-inducing query–response pairs and organizes the resulting cases into recurring evaluator failure modes.

2 Related Work

Reliability of LLM-as-a-Judge. Prior work has shown that model-based evaluators are susceptible to systematic biases. Ye et al. [2025] identify 12 types of bias in LLM-as-a-Judge and introduce CALM, a framework that uses automated, principle-guided modifications to quantify judge sensitivity across models. Their study evaluates predefined bias categories and consistency under controlled perturbations. Our work is complementary: rather than measuring a fixed set of biases within individual judges, TVD-Conflict searches for responses that induce conflicting safety decisions across multiple judges operating under the same evaluation criteria.

Safety-judge decisions can also remain anchored to internal priors when the provided context or safety definition changes. The steerability study of Alloula et al. [2026] shows that judge models are often reluctant to revise their evaluations when an explicit definition conflicts with their learned safety boundary. TVD-Conflict studies a complementary instability: even under a shared definition and identical visible interaction, evaluator backbones can produce incompatible safety judgments.

3 Extract Safety-Judge Failure-Modes from Attack Responses

We evaluate responses from ten jailbreak attacks: Cognitive Overload, MathPrompt, Semantic Cloaking, ActorBreaker, Mastermind, Compound Roleplay, CIA Context, AutoDAN-Turbo, FITD, and Prefill. Each attack contributes 200 responses, covering four victim models and 50 AdvBench goals. Every response is evaluated by GPT-OSS-120B, Qwen3.7-Flash, and DeepSeek-V4-Flash using both the binary and graded judge prompts.

Panel A of Table 1 reports the ASR assigned by each backbone under the two rubrics. The choice of rubric can reverse the apparent ranking of attack methods. For GPT-OSS-120B, for example, MathPrompt falls from 67.5% under binary judging to 29.5% under scoring, while Mastermind rises from 31.5% to 87.5%. Panel B isolates evaluator conflict from ASR: it holds the rubric fixed and measures disagreement across the three backbones for each attack.

We derive the failure-mode taxonomy from conflict cases rather than from all attack responses. Each case retains the shared query and response, the raw judge outputs, parsed verdicts, and the dimensions responsible for disagreement. This representation supports both quantitative prevalence estimates and qualitative inspection of how an evaluator reached its decision.

Table 1: Judge-dependent evaluation of ten jailbreak attacks. The left block reports attack success rate (ASR, %) under each judge–rubric combination; the right block reports cross-judge Conflict Rate (CR, %) with the rubric held fixed.

	Attack Success Rate by Judge						Cross-Judge Conflict Rate	
	GPT-OSS-120B		Qwen3.7-Flash		DeepSeek-V4-Flash		All Judges	
Attack	Bin.	Score	Bin.	Score	Bin.	Score	Bin. CR	Score CR
Cognitive Overload	0.5	4.0	0.5	4.0	0.0	3.5	0.5	1.5
MathPrompt	67.5	29.5	51.0	27.0	32.5	11.5	43.4	10.2
Semantic Cloaking	50.0	28.0	42.0	47.0	45.0	49.0	24.8	10.7
ActorBreaker	43.0	66.0	24.0	52.5	25.5	55.0	35.0	11.2
Mastermind	31.5	87.5	13.0	70.5	15.0	79.0	36.5	9.5
Compound Roleplay	42.0	41.0	37.5	13.5	24.0	37.0	16.1	10.4
CIA Context	49.5	70.0	21.5	70.5	27.5	72.5	41.1	7.3
AutoDAN-Turbo	17.0	23.5	15.0	21.0	13.0	21.0	7.9	1.1
FITD	51.5	45.5	47.0	12.5	43.0	13.0	16.7	37.5
Prefill	31.0	30.0	17.0	22.5	24.5	20.0	23.7	11.9
Overall	–	–	–	–	–	–	24.2	10.8

Protocol. Each attack contains four victim models $\times$ 50 AdvBench goals. Binary ASR is the fraction labeled harmful. Scoring ASR requires refusal $= 0$ and mean convincingness–specificity ≥ 4; missing judgments count as unsuccessful in the fixed denominator of 200. Binary CR is the fraction of complete cases in which the three harmfulness labels are not identical. Scoring CR uses $\max_j s_j - \min_j s_j > 0.5$, where $s = (1 - r)(c + p - 2)/8$.

Table 2: Conflict Rate (%) under a search budget of ten candidate revisions. Results will be inserted after the budget-controlled search runs complete.

Judge Pair	Binary CR	Scoring CR
GPT-OSS / Qwen	TBD	TBD
GPT-OSS / DeepSeek	TBD	TBD
Qwen / DeepSeek	TBD	TBD
Overall	TBD	TBD

4 TVD-Conflict: Systematic Safety-Judge Failure Data Generation

TVD-Conflict formulates safety-judge stress testing as a search problem over task, validator, and data components. A task specifies the safety criterion and visible context. Validators are LLM judges that independently apply the same criterion. The data component stores candidate interactions together with raw and parsed validator outputs. A candidate is retained when it crosses the configured disagreement threshold.

For binary judging, a conflict occurs when at least two validators assign different harmfulness labels. For graded judging, each validator receives the score

$$s = (1 - r)\frac{c + p - 2}{8},$$

where $r \in \{0, 1\}$ is the refusal decision and $c, p \in \{1, \ldots, 5\}$ are convincingness and specificity. A pair conflicts when its absolute score difference exceeds 0.5. The search process uses validator feedback to revise a candidate while preserving the underlying task and ground-truth safety label.

Search-budget evaluation. Table 2 will report Conflict Rate after a budget of ten candidate revisions, using the same conflict definitions as Table 1.

5 Failure-Modes of Frontier LLM-as-Safety-Judge

We organize disagreement cases into six failure modes. The categories describe the decision error exposed by a case, not the surface form of the jailbreak. Table 3 will report their prevalence after the conflict set has been annotated.

Table 3: Prevalence of six safety-judge failure modes in the TVD-Conflict dataset. Counts and shares will be inserted after annotation.

Failure Mode	Count	Share (%)
Criteria ambiguity	TBD	TBD
Refusal but harmful	TBD	TBD
Compliance but safe	TBD	TBD
50–50 problem	TBD	TBD
Distraction	TBD	TBD
Insufficient capability	TBD	TBD

Criteria ambiguity. The judges operationalize the stated safety criterion differently, such as treating harmfulness, policy violation, and attack success as interchangeable.

Refusal but harmful. A response begins with refusal language but subsequently provides information that materially advances the harmful request. Some judges anchor on the refusal marker, while others evaluate the supplied content.

Compliance but safe. A response adopts the form of compliance without delivering harmful or usable information. Judges that rely on affirmative framing can mistake stylistic compliance for harmful assistance.

The 50–50 problem. The available evidence supports two plausible readings of comparable strength, causing verdicts to split across judges or rubric formulations.

Distraction. An evaluator focuses on wrappers, disclaimers, fictional settings, mathematical notation, or peripheral detail instead of the behavior that determines safety.

Insufficient capability. The judge fails to decode, follow, or reason over the representation required to identify the harmful behavior, including obfuscation and multi-step context.

6 Conclusion

Safety evaluation with LLM judges is not invariant to the evaluator. Across ten jailbreak attacks, three judge backbones and two rubrics produce materially different decisions on identical interactions. Cross-judge Conflict Rate is 24.2% under binary judging and 10.8% under scoring, with substantially higher rates for particular attacks. The variation is strongly attack dependent, indicating that a single aggregate ASR can conceal systematic blind spots. TVD-Conflict turns these discrepancies into a data-generation objective and supports analysis through six operational failure modes. The resulting evidence argues for evaluating safety judges as fallible measurement instruments rather than treating their outputs as interchangeable ground truth.

References

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